

Aufgabe 1: (16 Punkte)

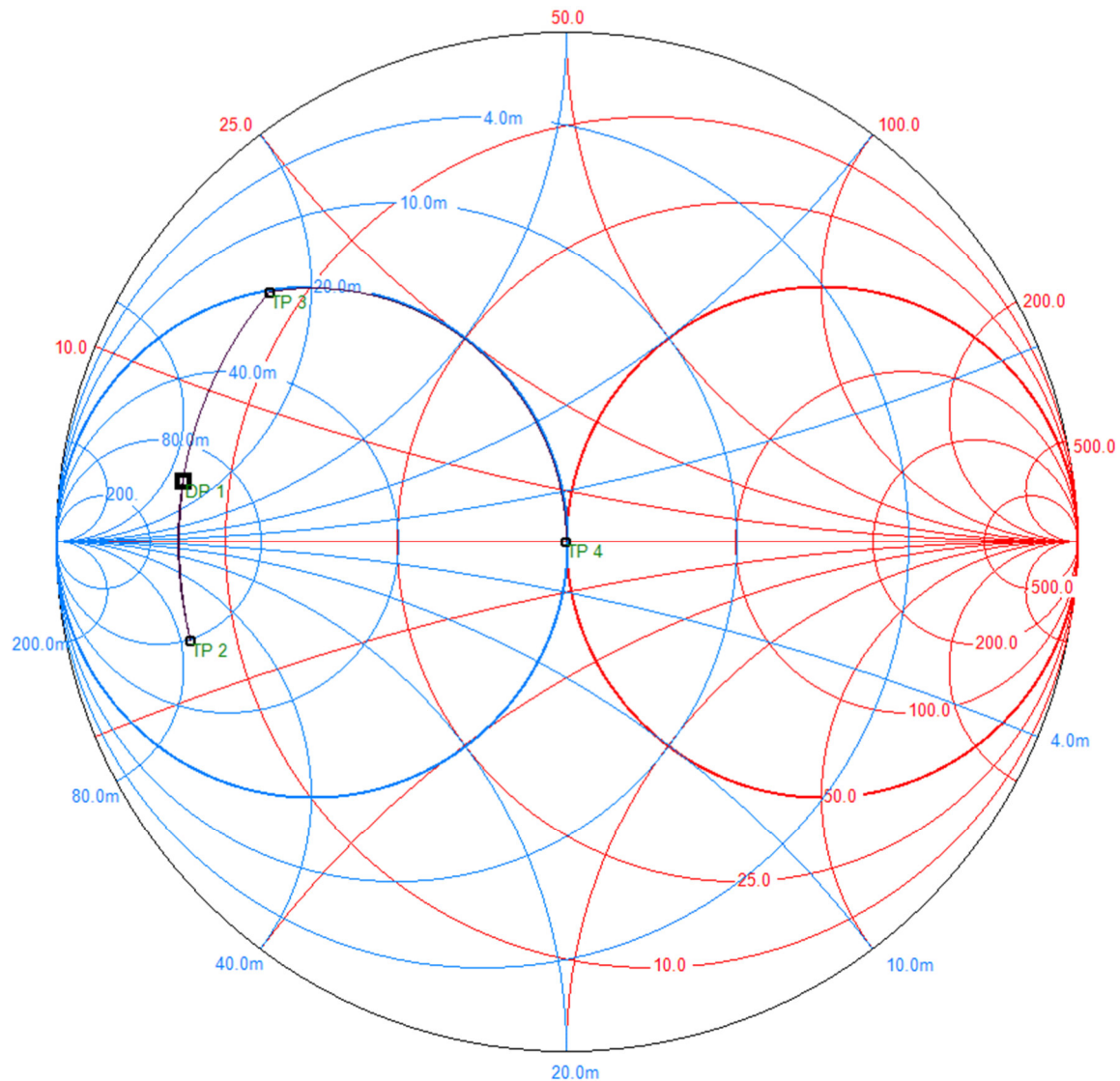
1.

$$\underline{Z}_{e,Tr} = Z_L \cdot \frac{1 + \underline{s}_{11}}{1 - \underline{s}_{11}} = 50\Omega \cdot \frac{1 + 0,762 \cdot e^{j171^\circ}}{1 - 0,762 \cdot e^{j171^\circ}} = (6,79 + j3,86)\Omega$$

$$\underline{Z}_{e,Tr} \text{ normiert: } \frac{\underline{Z}_{e,Tr}}{Z_L} = \frac{(6,79 + j3,86)\Omega}{50\Omega} = 0,136 + j0,0773$$

$$x_{C_1} = -\frac{1}{\omega \cdot C_1 \cdot Z_L} = -\frac{1}{2\pi \cdot 775 \cdot 10^6 \frac{1}{s} \cdot 20 \cdot 10^{-12} \frac{As}{V} \cdot 50 \frac{V}{A}} = -0,205$$

$\underline{Z}_{e,Tr}$ ins Smith-Chart eintragen (oder $\underline{s}_{11}$ mit Betrag und Phase direkt ins Smith Chart) und x_{C1} dransetzen. Kreis um Ursprung bis Schnittpunkt mit Leitwertkreis durch Ursprung ergibt l_1/λ . Weiter entlang des Kreises in den Ursprung ergibt b_{C2} .



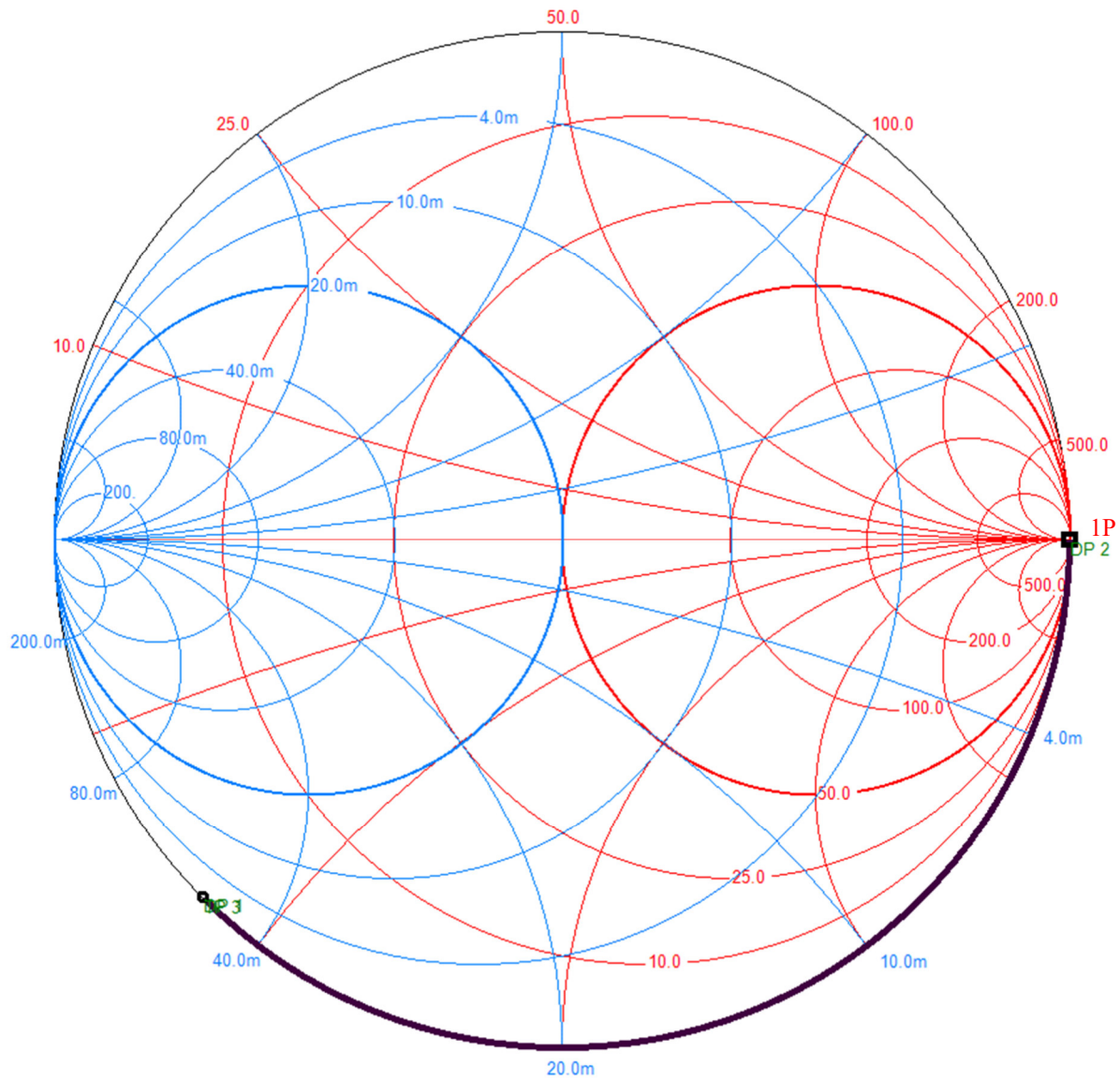
ablesen: $l_1/\lambda = 0,0555 - 0,4792 (+ 0,50) = 0,0763$ $b_{C2} = 2,38$

$$\lambda = \frac{c_0}{\sqrt{\epsilon_{r,eff}} \cdot f_0} = \frac{3 \cdot 10^8 \frac{m}{s}}{\sqrt{4,5} \cdot 775 \cdot 10^6 \frac{1}{s}} = 182,5 \text{ mm} \quad \underline{\underline{l_1 = 0,0763 \cdot 182,5 \text{ mm} = 13,9 \text{ mm}}}$$

$$\underline{\underline{C_2 = \frac{b_{C2}}{Z_L \cdot \omega} = \frac{2,38}{50 \frac{V}{A} \cdot 2\pi \cdot 775 \cdot 10^6 \frac{1}{s}} = 9,8 \text{ pF}}}$$

2. $b_{C_2} = 2,38 \quad \left[b_{C_2} = Z_L \cdot \omega \cdot C_2 = 50 \frac{\text{V}}{\text{A}} \cdot 2\pi \cdot 775 \cdot 10^6 \frac{1}{\text{s}} \cdot 10 \cdot 10^{-12} \frac{\text{As}}{\text{V}} = 2,43 \right]$

Leerlauf im Smith-Chart über Längsleitung in b_{C_2} transformieren:



ablesen: $l_2/\lambda = 0,437 \quad [0,457]$

$$\underline{\underline{l_s = (0,437 - 0,25) \cdot \lambda = 0,187 \cdot 182,5\text{mm} = 34,1\text{mm}}} \quad [34,3\text{mm}]$$

Aufgabe 2:

(11 Punkte)

1. mit 3-6 für $Z_L = 50\Omega$ und $\varepsilon_r = 4,5$: $\frac{w}{h} = 1,85 \rightarrow \underline{w} = 1,85 \cdot h = 1,85 \cdot 0,8\text{mm} = \underline{\underline{1,5\text{mm}}}$

mit 3-5 für $\frac{w}{h} = 1,85$ und $\varepsilon_r = 4,5$: $\varepsilon_{r,eff} = 3,3$

$$\underline{\underline{\lambda}} = \frac{c_0}{\sqrt{\varepsilon_{r,eff}} \cdot f_0} = \frac{3 \cdot 10^8 \frac{\text{m}}{\text{s}}}{\sqrt{3,3} \cdot 3,5 \cdot 10^9 \frac{1}{\text{s}}} = \underline{\underline{47,2\text{mm}}}$$

2.

$$\underline{s}_{41} = \frac{jk_L \sin(\beta l)}{\sqrt{1 - k_L^2} \cdot \cos(\beta l) + j \sin(\beta l)}; \quad |\underline{s}_{41}|^2 = \frac{k_L^2 \sin^2(\beta l)}{(1 - k_L^2) \cdot \cos^2(\beta l) + \sin^2(\beta l)} = \frac{k_L^2 \sin^2(\beta l)}{1 - k_L^2 \cdot \cos^2(\beta l)}$$

$$|\underline{s}_{41}|^2 - |\underline{s}_{41}|^2 \cdot k_L^2 \cdot \cos^2(\beta l) = k_L^2 \sin^2(\beta l); \quad |\underline{s}_{41}|^2 = |\underline{s}_{41}|^2 \cdot k_L^2 \cdot \cos^2(\beta l) + k_L^2 \sin^2(\beta l)$$

$$k_L^2 = \frac{|\underline{s}_{41}|^2}{|\underline{s}_{41}|^2 \cdot \cos^2(\beta l) + \sin^2(\beta l)}$$

da $|\underline{s}_{41}|^2 \ll 1$: $k_L^2 \approx \frac{|\underline{s}_{41}|^2}{\sin^2\left(\frac{2\pi}{\lambda} \cdot l\right)} = \frac{0,01}{\sin^2\left(\frac{2\pi}{47,2\text{mm}} \cdot 20\text{mm}\right)} = 4,704 \cdot 10^{-2} \quad [2,894 \cdot 10^{-2}]$

$$k_L = 0,217 \quad [0,170]$$

genau:

$$k_L^2 = \frac{|\underline{s}_{41}|^2}{|\underline{s}_{41}|^2 \cdot \cos^2(\beta l) + \sin^2(\beta l)} = \frac{|\underline{s}_{41}|^2}{|\underline{s}_{41}|^2 \cdot \cos^2\left(\frac{2\pi}{\lambda} \cdot l\right) + \sin^2\left(\frac{2\pi}{\lambda} \cdot l\right)} =$$

$$= \frac{0,01}{0,01 \cdot \cos^2\left(\frac{2\pi}{47,2\text{mm}} \cdot 20\text{mm}\right) + \sin^2\left(\frac{2\pi}{47,2\text{mm}} \cdot 20\text{mm}\right)} = 4,536 \cdot 10^{-2} \quad [2,841 \cdot 10^{-2}]$$

$$k_L = 0,213 \quad [0,169]$$

mit 7-12: $\frac{s}{h} = 0,6 \quad [0,8] \quad \underline{\underline{s}} = 0,6 \cdot h = 0,6 \cdot 0,8\text{mm} = \underline{\underline{0,48\text{mm}}} \quad [0,64\text{mm}]$

Aufgabe 3:

(10 Punkte)

1.

$$\begin{aligned}
 K &= \frac{1 - |\underline{s}_{11}|^2 - |\underline{s}_{22}|^2 + |\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}|^2}{2|\underline{s}_{12}\underline{s}_{21}|} = \\
 &= \frac{1 - 0,754^2 - 0,303^2 + |0,754 \cdot e^{-j171^\circ} \cdot 0,303 \cdot e^{-j53^\circ} - 0,0271 \cdot e^{j35^\circ} \cdot 9,10 \cdot e^{j38^\circ}|^2}{2 \cdot 0,0271 \cdot 9,10} = \\
 &= 0,814 < 1 \quad \rightarrow \underline{\underline{\text{nicht absolut stabil}}}
 \end{aligned}$$

$$\left[\begin{aligned}
 B &= 1 + |\underline{s}_{11}|^2 - |\underline{s}_{22}|^2 - |\underline{s}_{11}\underline{s}_{22} - \underline{s}_{12}\underline{s}_{21}|^2 = \\
 &= 1 + 0,754^2 - 0,303^2 - |0,754 \cdot e^{-j171^\circ} \cdot 0,303 \cdot e^{-j53^\circ} - 0,0271 \cdot e^{j35^\circ} \cdot 9,10 \cdot e^{j38^\circ}|^2 = \\
 &= 1,41 > 0 \quad \text{"stabil"}
 \end{aligned} \right]$$

2.

$$\underline{\underline{v_{Tr}}} = 10\text{dB} \cdot \lg |\underline{s}_{21}|^2 = 10\text{dB} \cdot \lg(9,10^2) = \underline{\underline{19,2\text{dB}}}$$

3.

$$\begin{aligned}
 \underline{\underline{v_{zUS}}} &= 10\text{dB} \cdot \lg \left(\frac{1}{1 - |\underline{s}_{11}|^2} \cdot \frac{1}{1 - |\underline{s}_{22}|^2} \right) = 10\text{dB} \cdot \lg \left(\frac{1}{1 - 0,754^2} \cdot \frac{1}{1 - 0,303^2} \right) = \\
 &= 10\text{dB} \cdot \lg(2,552) = \underline{\underline{4,1\text{dB}}}
 \end{aligned}$$

4.

$$\begin{aligned}
 \underline{Z_{a,Tr}} &= Z_L \cdot \frac{1 + \underline{s}_{22}}{1 - \underline{s}_{22}} = 50\Omega \cdot \frac{1 + 0,303 \cdot e^{-j53^\circ}}{1 - 0,303 \cdot e^{-j53^\circ}} = 70,8\Omega \cdot e^{-j28^\circ} \\
 \underline{\underline{Z_{Last}}} &= \underline{Z_{a,Tr}}^* = \underline{\underline{70,8\Omega \cdot e^{+j28^\circ}}}
 \end{aligned}$$

Aufgabe 4:

(13 Punkte)

1.

$$F_{VV1} = 10^{\frac{f_{VV1}}{10\text{dB}}} = 10^{\frac{1,2\text{dB}}{10\text{dB}}} = 1,32 \quad G_{VV1} = 10^{\frac{g_{VV1}}{10\text{dB}}} = 10^{\frac{6,0\text{dB}}{10\text{dB}}} = 3,98$$

$$F_{VV2} = 10^{\frac{f_{VV2}}{10\text{dB}}} = 10^{\frac{2,0\text{dB}}{10\text{dB}}} = 1,58 \quad G_{VV2} = 10^{\frac{g_{VV2}}{10\text{dB}}} = 10^{\frac{20\text{dB}}{10\text{dB}}} = 100$$

$$F_L = 10^{\frac{a_L}{10\text{dB}}} = 10^{\frac{10\text{dB}}{10\text{dB}}} = 10 \quad G_L = 10^{\frac{a_L}{-10\text{dB}}} = 10^{\frac{10\text{dB}}{10\text{dB}}} = 0,1$$

$$F_{ges} = 1 + (F_{VV1} - 1) + \frac{F_L - 1}{G_{VV1}} + \frac{F_{VV2} - 1}{G_{VV1} \cdot G_L} = 1 + (1,32 - 1) + \frac{10 - 1}{3,98} + \frac{1,58 - 1}{3,98 \cdot 0,1} = 5,04$$

$$\underline{\underline{f_{ges} = 10\text{dB} \cdot \lg F_{ges} = 7,02\text{dB}}}$$

$$F_{ges2} = 1 + (F_{VV1} - 1) + \frac{F_{VV2} - 1}{G_{VV1}} + \frac{F_L - 1}{G_{VV1} \cdot G_{VV2}} = 1 + (1,32 - 1) + \frac{1,58 - 1}{3,98} + \frac{10 - 1}{3,98 \cdot 100} = 1,49$$

$$\underline{\underline{f_{ges2} = 10\text{dB} \cdot \lg F_{ges} = 1,72\text{dB}}}$$

2.

$$\left(\frac{S}{N}\right)_{Ant} = F_{ges} \cdot \left(\frac{S}{N}\right)_{Empf,min} \triangleq 7,0\text{dB} + 15\text{dB} = 22\text{dB}$$

$$P_{RA} = k \cdot T_A \cdot B = 1,38 \cdot 10^{-23} \frac{\text{Ws}}{\text{K}} \cdot 10\text{K} \cdot 500 \cdot 10^6 \frac{1}{\text{s}} = 6,9 \cdot 10^{-14} \text{W} \triangleq -101,6\text{dBm}$$

$$\underline{\underline{P_{SA} = -101,6\text{dBm} + 22\text{dB} = -79,6\text{dBm}}}$$

Aufgabe 5:

(17 Punkte)

1.

$$\underline{r}_L = \frac{R_L - Z_L}{R_L + Z_L} = \frac{25\Omega - 50\Omega}{25\Omega + 50\Omega} = -\frac{1}{3}$$

$$\underline{b}_{Q2} = \underline{b}_Q \cdot \frac{\underline{s}_{21}}{1 - \underline{s}_{11} \cdot \underline{r}_Q} = \underline{b}_Q \cdot \frac{j0,93}{1 - 0,36 \cdot \frac{2}{3}} = \underline{b}_Q \cdot j1,2237$$

$$\underline{r}_2 = \underline{s}_{22} + \frac{\underline{s}_{12} \cdot \underline{s}_{21} \cdot \underline{r}_Q}{1 - \underline{s}_{11} \cdot \underline{r}_Q} = 0,36 + \frac{j0,93 \cdot j0,93 \cdot \frac{2}{3}}{1 - 0,36 \cdot \frac{2}{3}} = -0,3987$$

$$\underline{\underline{P_L}} = \frac{1}{2} |\underline{b}_{Q2}|^2 \cdot \frac{1 - |\underline{r}_L|^2}{|1 - \underline{r}_2 \cdot \underline{r}_L|^2} = \frac{1}{2} |\underline{b}_Q|^2 \cdot 1,2237^2 \cdot \frac{1 - \left|-\frac{1}{3}\right|^2}{|1 - (-0,3987) \cdot \left(-\frac{1}{3}\right)|^2} = 5\text{W} \cdot 1,770 = \underline{\underline{8,852\text{W}}}$$

2. Last direkt an Quelle:

$$P_{L,abs0} = P_L \cdot \frac{1 - |\underline{r}_L|^2}{|1 - \underline{r}_Q \cdot \underline{r}_L|^2} = 5,0\text{W} \cdot \frac{1 - \left|-\frac{1}{3}\right|^2}{\left|1 - \frac{2}{3} \cdot \left(-\frac{1}{3}\right)\right|^2} = 5,0\text{W} \cdot 0,5950 = 2,975\text{W}$$

$$\underline{\underline{a_e}} = 10\text{dB} \cdot \lg\left(\frac{P_{L,abs0}}{P_L}\right) = 10\text{dB} \cdot \lg\left(\frac{2,975\text{W}}{8,852\text{W}}\right) = \underline{\underline{-4,7\text{dB}}} \quad [-3,7\text{dB}]$$

3.

$$\underline{r}_1 = \underline{s}_{11} + \frac{\underline{s}_{12} \cdot \underline{s}_{21} \cdot \underline{r}_L}{1 - \underline{s}_{22} \cdot \underline{r}_L} = 0,36 + \frac{j0,93 \cdot j0,93 \cdot \left(-\frac{1}{3}\right)}{1 - 0,36 \cdot \left(-\frac{1}{3}\right)} = 0,6174$$

$$P_1 = P_Q \cdot \frac{1 - |\underline{r}_1|^2}{|1 - \underline{r}_Q \cdot \underline{r}_1|^2} = 5,0\text{W} \cdot \frac{1 - 0,6174^2}{\left|1 - \frac{2}{3} \cdot 0,6174\right|^2} = 5,0\text{W} \cdot 1,787 = 8,937\text{W}$$

$$\underline{\underline{a_b}} = 10\text{dB} \cdot \lg\left(\frac{P_1}{P_L}\right) = 10\text{dB} \cdot \lg\left(\frac{8,937\text{W}}{8,852\text{W}}\right) = \underline{\underline{0,042\text{dB}}} \quad [1,06\text{dB}]$$